Help U WhatsApp-to-Drive Bookkeeper

🏗️ 1. Core Architecture (The "Serverless" Blueprint)

- **Interface:** WhatsApp Cloud API (Meta).
- **The Brain (Logic):** n8n (Self-hosted on a $6 Droplet) or Google Apps Script (Free).
- **The Intelligence:** OpenAI GPT-4o-mini (Vision for bills, NLP for voice).
- **The Database/Storage:** User-Owned Google Drive (via OAuth2 `drive.file` scope).

🛠️ 2. Functional Modules & Workflows

Module A: Smart Purchase Capture

- **Input:** User snaps a photo of a physical bill in the WhatsApp chat.
- **AI Logic:** Vision API extracts: *Vendor GSTIN, Date, HSN Codes, Taxable Value, CGST/SGST/IGST.*
- **Output:** 1. Uploads image to `UserDrive/Help U/Purchases/Feb_2026/`. 2. Appends data row to `UserDrive/Help U/Master_Ledger.gsheet`.

Module B: Conversational Sales Invoicing

- **Input:** Voice note or Text: "*Bill for Apollo Pharm, 20 units Masks at 150 each, 12% GST.*"
- **AI Logic:** Parses customer, quantity, and price. Calculates GST math.
- **Output:** 1. Fills a Google Doc template in the User's Drive. 2. Converts to PDF. 3. Sends PDF link back to user on WhatsApp.

Module C: GST Portal Ready Reports

- **Trigger:** Monthly alert sent by the bot on the 5th of the month.
- **Generation:** Script reads the `Master_Ledger` sheet.
- **Output:** Generates a **CSV/JSON** file formatted exactly for the **GSTR-1 Offline Tool**.
- **Filing:** User downloads from WhatsApp and uploads to `gst.gov.in`.

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🚀 5. Implementation Roadmap (4 Days)

- Day **1:** Setup Meta Dev Account & Google Cloud Console (OAuth2).
- Day **2:** Build the n8n "Purchase Scan" workflow (WhatsApp → AI → Google Sheet).
- Day **3:** Build the "Sales Prompt" workflow (Voice → Doc Template → PDF).
- Day **4:** Build the GST Export script & conduct pilot testing with 5 local traders.

The AI System Prompt (The "Brain")

Copy and paste this into your AI configuration (OpenAI GPT-4o-mini). This ensures the AI understands Indian GST rules and extracts data perfectly.

Role: You are a specialist Indian GST Compliance Accountant. **Task:** Analyze the provided image (bill) or text/voice transcript (sale) and extract data into a strict JSON format.

Extraction Rules:

1. **Identify GSTINs:** Find both 'Supplier GSTIN' and 'Recipient GSTIN'.
2. **Tax Split:** Calculate or extract CGST, SGST, and IGST separately. If the state of the supplier matches the user, use CGST/SGST. If different, use IGST.
3. **HSN Codes:** Extract the 4 or 8-digit HSN code. If missing, suggest the most likely 4-digit code based on the item description.
4. **Invoice Type:** > - Set to `B2B` if a Recipient GSTIN is present.
    - Set to `B2C` if no Recipient GSTIN is found.

Required JSON Output Format:

JSON

```
{
  "transaction_type": "Purchase/Sale",
  "invoice_no": "string",
  "date": "DD-MM-YYYY",
  "vendor_name": "string",
  "vendor_gstin": "string",
  "hsn_code": "string",
  "taxable_value": 0.00,
  "gst_rate": 0,
```

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```
  "cgst": 0.00,
  "sgst": 0.00,
  "igst": 0.00,
  "total_amount": 0.00
}
```

🛠️ Technical Architecture & Developer Plan

Step 1: The "Entry Point" (WhatsApp)

- **Tool:** WhatsApp Cloud API (Meta).
- **Setup:** Create a Webhook that points to your serverless backend. When a user sends a message, Meta sends the "Media ID" (image) or "Audio ID" (voice) to your script.

Step 2: The "Brain" (n8n or Python)

- **Action:** The script takes the Media ID, downloads the file from Meta's servers, and passes it to **OpenAI Vision API**.
- **Voice Handling:** For voice prompts, use **OpenAI Whisper** to convert the audio to text before sending it to the "Brain."

Step 3: The "Database" (Google Sheets & Docs)

- **Permission:** Use **Google OAuth2** with the `https://www.googleapis.com/auth/drive.file` scope.
- **Operation:** - **Purchases:** Append a row to a Google Sheet in the user's Drive.
  - **Sales:** Fill a "Google Doc Template" (pre-created in their drive) and export as PDF.
  - **Monthly Report:** At the end of the month, a script reads the Sheet and creates a **CSV file** with headers matching the **GSTR-1 Offline Tool**.

📊 Monthly GST Report Column Headers

Ensure your Google Sheet uses these exact headers so the monthly export is "Portal Ready":

GSTIN/ UIN of Recipient	Receiver Name	Invoice Number	Invoice date	Invoice Value	Place Of Supply	Reverse Charge	Applicable % of Tax Rate	Invoice Type	E-Commerce GSTIN	Rate	Taxable Value	Cess Amount

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📅 Your Execution Checklist

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1. **Meta Dev Account:** Register your business phone number on developers.facebook.com.
2. **Google Cloud:** Create a project, enable "Google Drive API" and "Google Sheets API," and set up the OAuth screen.
3. **The Template:** Create one sample "Invoice Template" in Google Docs and one "Master Ledger" in Google Sheets.
4. **The Pilot:** Test with 10 actual paper bills to see if the AI identifies the small, blurry text on thermal receipts.

key to generating a central report from data that "lives" with the user is **On-Demand Aggregation**. You don't store their data permanently; you "fetch and forget" it only when the report is needed.

Since your architecture uses **n8n** or **Google Apps Script** as the "Brain," the process is highly efficient and serverless.

1. The "Fetch & Forget" Report Workflow

Instead of keeping a copy of every user's sheet on your server, your system acts as a temporary bridge.

1. **Trigger:** On the 1st of each month, your central serverless script (n8n) wakes up.
2. **Auth Loop:** The script iterates through your list of registered User IDs and their stored **OAuth Refresh Tokens**.
3. **Data Pull:** Using the token, the script calls the Google Sheets API to "Read" only the rows from the last 30 days in that user's specific `Master_Ledger`.
4. **Local Processing (RAM Only):** The data is held in the script's temporary memory (RAM). It reformats the data into the **GSTR-1 CSV/JSON** structure.
5. **Delivery:** The script "pushes" the completed file back to the user via WhatsApp or saves it into their Drive.
6. **Cleanup:** The script finishes and wipes its temporary memory. You never "held" the data; you just processed it.

2. Making the File "Portal Ready"

The GST portal (`gst.gov.in`) is strict. Your aggregation script must perform these three "Cleanup" tasks during the report generation:

- **Validation:** Check if any GSTINs are missing digits.
- **HSN Mapping:** Ensure every row has a valid 4 or 8-digit HSN code (using your AI to fill gaps).

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- **Format Transformation:** Convert the Google Sheet rows into the specific columns required by the **GST Offline Tool**.

Your Sheet Column	GST Portal Requirement
`Vendor Name`	Not needed (GSTIN is enough).
`Total Amount`	Must be split into `Taxable Value` and `Tax Amount`.
`State`	Must be converted to a 2-digit **State Code** (e.g., Telangana is `36`).

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3. Developer Implementation Plan

To do this cost-effectively, tell your developers to use **n8n's "Google Sheets Trigger"** or a **"Schedule Trigger"**.

- **The "User List" Sheet:** You keep one private Google Sheet that lists: `User_Phone_Number`, `User_Google_Drive_Folder_ID`, and `OAuth_Refresh_Token`.
- **The Aggregator Node:** This node loops through the list, enters each user's sheet, grabs the data, and uses a **"Function Node" (JavaScript)** to create the CSV.
- **No Database Cost:** Because the "User List" is just another Google Sheet, you have **zero database hosting fees**.

* ***This idea is completely drafted by Sriram Prasad Munnangi , and all Intellectual rights belong to Sriram Prasad Munnangi**

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